

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12389116
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Inagaki; Yu

Focus control apparatus, image pickup apparatus, and focus control method

Abstract

A focus control apparatus preforms a focus detection using a phase-difference detecting method, and controls a drive of a focus lens included in an optical system based on a focus detection result acquired by the focus detection. The focus control apparatus sets, in a search operation of acquiring the focus detection result while driving the focus lens in a search direction, a movable range of the focus lens based on the search direction and a position of the focus lens, in a case where the focus detection result for a position within the movable range is acquired, drives the focus lens based on the focus detection result, and in a case where the focus detection result for a position outside the movable range is acquired, drives the focus lens in the search direction without using the focus detection result.

Inventors:	Inagaki; Yu (Tokyo, JP)
Applicant:	CANON KABUSHIKI KAISHA (Tokyo, JP)
Family ID:	91079629
Assignee:	CANON KABUSHIKI KAISHA (Tokyo, JP)
Appl. No.:	18/498433
Filed:	October 31, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240171858 A1	May. 23, 2024

Foreign Application Priority Data

JP	2022-184290	Nov. 17, 2022
----	-------------	---------------

Publication Classification

Int. Cl.: H04N23/67 (20230101)

U.S. Cl.:

CPC H04N23/675 (20230101);

Field of Classification Search

CPC: H04N (23/672); H04N (23/673); H04N (23/675)

USPC: 348/345

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10477094	12/2018	Maruyama	N/A	H04N 23/67
10547779	12/2019	Wan	N/A	H04N 23/672
11012609	12/2020	Shimizu	N/A	H04N 23/675
11490000	12/2021	Feng	N/A	H04N 23/672
11877051	12/2023	Inagaki	N/A	H04N 23/611
2013/0182171	12/2012	Kawanishi	348/345	H04N 23/663
2015/0323760	12/2014	Tomosada	250/201.2	G02B 7/28
2016/0295101	12/2015	Ogino	N/A	H04N 23/611
2016/0301854	12/2015	Kato	N/A	G02B 15/143
2017/0289438	12/2016	Maruyama	N/A	G03B 13/36
2017/0343768	12/2016	Shirai	N/A	G03B 13/36
2018/0081148	12/2017	Tomita	N/A	G03B 17/14
2018/0213172	12/2017	Ohtsubo	N/A	G02B 7/34
2019/0335089	12/2018	Kadambala	N/A	G06V 10/25
2020/0228719	12/2019	Kimoto	N/A	H04N 23/69
2020/0271765	12/2019	Glover	N/A	G01S 7/4876
2022/0116545	12/2021	Feng	N/A	H04N 23/80
2022/0353431	12/2021	Inagaki	N/A	H04N 23/611

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2007-164051	12/2006	JP	N/A

Primary Examiner: Prabhakher; Pritham D

Attorney, Agent or Firm: Cowan, Liebowitz & Latman, P.C.

Background/Summary

BACKGROUND

Technical Field

(1) One of the aspects of the embodiments relates to a focus adjustment (focus) control.

Description of Related Art

(2) Some image pickup apparatuses perform a search operation in which a focus lens is moved to search for an in-focus position. Japanese Patent Laid-Open No. 2007-164051 discloses a method of performing the search operation limited to a far side of a near side endpoint or a near side of a far side endpoint, respectively, according to a user's designation of a current lens position as the near side endpoint or the far side endpoint as a starting position for the search operation.

(3) However, the method disclosed in Japanese Patent Laid-Open No. 2007-164051 is based on a focus adjustment control using a contrast detecting method. If the method in Japanese Patent Laid-Open No. 2007-164051 is applied to a focus adjustment control using a phase-difference detecting method, there is a risk that near a start position of the search operation, a defocus amount near the start point is detected and the start position of the search operation is refocused based on the defocus amount. As a result, it is not possible to perform an appropriate search operation to quickly focus on an object desired by the user.

SUMMARY

(4) A focus control apparatus according to one aspect of the embodiment includes a memory configured to store instructions, and a processor configured to execute the instructions, the processor being configured to preform a focus detection using a phase-difference detecting method, and control a drive of a focus lens included in an optical system based on a focus detection result acquired by the focus detection. The processor sets, in a search operation of acquiring the focus detection result while driving the focus lens in a search direction, a movable range of the focus lens based on the search direction and a position of the focus lens, in a case where the focus detection result for a position within the movable range is acquired, drives the focus lens based on the focus detection result, and in a case where the focus detection result for a position outside the movable range is acquired, drives the focus lens in the search direction without using the focus detection result.

(5) Further features of the disclosure will become apparent from the following description of embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram illustrating a configuration of an image pickup apparatus according to a first embodiment.

(2) FIG. 2 is a diagram illustrating a pixel array of an image sensor according to the first embodiment.

(3) FIGS. 3A and 3B are a plan view and a cross-sectional view of a pixel according to the first embodiment, respectively.

(4) FIG. 4 is a diagram for explaining a pupil division according to the first embodiment.

(5) FIG. 5 is a diagram for explaining a relationship between the image sensor and the pupil division according to the first embodiment.

(6) FIG. 6 is a diagram for explaining a relationship between a defocus amount and an image shift amount according to the first embodiment.

(7) FIG. 7 is a flowchart illustrating an image pickup process according to the first embodiment.

(8) FIG. 8 is a flowchart illustrating a search AF process according to the first embodiment.

(9) FIG. 9 is a flowchart illustrating a process of calculating a movable range of a focus lens according to the first embodiment.

- (10) FIG. **10** is a diagram illustrating a positional relationship between an object and a background according to the first embodiment.
- (11) FIGS. **11A** and **11B** are diagrams illustrating signals of an object and a background according to the first embodiment.
- (12) FIG. **12** is a diagram illustrating a relationship between a focus lens position and a defocus amount according to the first embodiment.
- (13) FIGS. **13A**, **13B**, **13C**, and **13D** are diagrams each illustrating a relationship among a search direction, a search start position, a focus lens position, a movable range of a focus lens, a defocus amount, and a focus lens drive amount according to the first embodiment.
- (14) FIG. **14** is a flowchart illustrating a search AF process according to a second embodiment.
- (15) FIGS. **15A**, **15B**, **15C**, and **15D** are diagrams each illustrating a relationship among a search direction, a search start position, a focus lens position, a defocus amount, and a focus lens drive amount according to the second embodiment.

DESCRIPTION OF THE EMBODIMENTS

(16) In the following, the term “unit” may refer to a software context, a hardware context, or a combination of software and hardware contexts. In the software context, the term “unit” refers to a functionality, an application, a software module, a function, a routine, a set of instructions, or a program that can be executed by a programmable processor such as a microprocessor, a central processing unit (CPU), or a specially designed programmable device or controller. A memory contains instructions or programs that, when executed by the CPU, cause the CPU to perform operations corresponding to units or functions. In the hardware context, the term “unit” refers to a hardware element, a circuit, an assembly, a physical structure, a system, a module, or a subsystem. Depending on the specific embodiment, the term “unit” may include mechanical, optical, or electrical components, or any combination of them. The term “unit” may include active (e.g., transistors) or passive (e.g., capacitor) components. The term “unit” may include semiconductor devices having a substrate and other layers of materials having various concentrations of conductivity. It may include a CPU or a programmable processor that can execute a program stored in a memory to perform specified functions. The term “unit” may include logic elements (e.g., AND, OR) implemented by transistor circuits or any other switching circuits. In the combination of software and hardware contexts, the term “unit” or “circuit” refers to any combination of the software and hardware contexts as described above. In addition, the term “element,” “assembly,” “component,” or “device” may also refer to “circuit” with or without integration with packaging materials.

(17) Referring now to the accompanying drawings, a detailed description will be given of embodiments according to the disclosure.

First Embodiment

(18) The image pickup system **10** according to the first embodiment shown in FIG. **1** is a lens-interchangeable single-lens reflex digital camera system that can perform autofocus using an imaging plane phase-difference detecting method (hereinafter referred to as imaging plane phase-difference AF). This embodiment and the second embodiment described later are applicable to a digital camera with an integrated lens and a digital video camera, and can also be applied to terminal devices such as tablets and smartphones, and various image pickup apparatuses such as monitoring cameras, car-mounted cameras, and medical cameras.

(19) <Apparatus Configuration>

(20) The image pickup system **10** consists of a lens unit **100** and a camera body **120** as an image pickup apparatus. The lens unit **100** is detachably connected to the camera body **120** via a mount **M**, indicated by a dotted line in the center of the figure.

(21) The lens unit **100** has an image pickup optical system that includes a first lens group **101**, a diaphragm **102**, a second lens group **103**, and a focus lens group (hereinafter referred to as focus lens) **104**.

(22) The first lens group **101** is disposed closest to an object in the lens unit **100** and is held so as to be movable in an optical axis direction OA. In the following, the optical axis direction OA is referred to as a Z direction, and a direction from a camera side to an object side is referred to as a positive direction. In this embodiment, an origin O of an axis in the Z direction corresponds to a position of an image sensor **122** in the camera body **120**, which will be described later.

(23) The diaphragm **102** adjusts a light amount by changing its opening diameter. The diaphragm **102** also functions as a mechanical shutter that controls an exposure time when a still image is captured. The diaphragm **102** and the second lens group **103** are movable in the optical axis direction OA as a single unit and move in conjunction with the first lens group **101** to achieve a zoom function.

(24) The focus lens **104** is movable in the optical axis direction OA, and an object distance at which the lens unit **100** is focused (in-focus distance) changes according to its position. In this embodiment, the autofocus is achieved by controlling a position of the focus lens **104** in the optical axis direction OA.

(25) The lens unit **100** has a drive/control system (including an equipment, a circuitry, a program code, and other items). Among the drive/control system, the drive system includes a zoom actuator **111**, a diaphragm/shutter actuator **112**, a focus actuator **113**, a zoom driver **114**, a diaphragm/shutter driver **115**, and a focus driver **116**. The control system that controls the drive system includes a lens MPU **117** and a lens memory **118**.

(26) The zoom actuator **111** drives the first lens group **101** and the second lens group **103** back and forth in the optical axis direction OA to perform a zoom control to change an angle of view of the image pickup optical system. The diaphragm/shutter actuator **112** controls the opening diameter of the diaphragm **102** to adjust the light amount, and controls an opening and closing movement of the diaphragm **102** to control the exposure time when an image is captured. The focus actuator **113** drives the focus lens **104** back and forth in the optical axis direction OA to perform the autofocus and also has a function of detecting a current position (actual position) of the focus lens **104**,

(27) The zoom driver **114** drives the zoom actuator **111** in response to a zoom operation by a user or a control value of the lens MPU **117**. The diaphragm/shutter driver **115** drives the diaphragm/shutter actuator **112**. The focus driver **116** drives the focus actuator **113**.

(28) The lens MPU **117** performs calculations related to the image pickup optical system and controls the zoom driver **114**, the diaphragm/shutter driver **115**, the focus driver **116**, and lens memory **118**. The lens MPU **117** can communicate commands and data with a camera MPU **125** via the mount M. For example, the lens MPU **117** detects the current position of the focus lens **104** and notifies the camera MPU **125** of lens position information in response to a request from the camera MPU **125**. The lens position information includes a position of the focus lens **104** in the optical axis direction OA, a position and diameter of an exit pupil in the optical axis direction OA, and a position and diameter of a lens frame that limits a light beam of the exit pupil in the optical axis direction OA.

(29) The lens MPU **117** also controls the zoom driver **114**, the diaphragm/shutter driver **115**, and the focus driver **116** in response to a request from the camera MPU **125**. The lens memory **118** stores in advance optical information necessary for the imaging plane phase-difference AF. The lens memory **118** also stores, for example, a defocus map that indicates a correspondence between a position or movement amount of the focus lens **104** and the defocus amount. The defocus map is generated by calculating the defocus amount at each pixel position of the image sensor **122**, which will be described later. When the lens MPU **117** receives a request from the camera MPU **125** to change the defocus amount by a predetermined amount, the lens MPU **117** refers to the defocus map stored in the lens memory **118**. The lens MPU **117** controls the focus actuator **113** to move the focus lens **104** by a distance corresponding to the predetermined amount.

(30) The camera MPU **125** controls operations of the lens unit **100** by executing programs stored in a ROM **125a** and the lens memory **118**. The lens memory **118** also stores optical information and

other information on the image pickup optical system.

(31) The camera body **120** has an optical low-pass filter **121**, the image sensor **122**, and a drive/control system described later. The optical low-pass filter **121** reduces false colors and moires in a captured image.

(32) The image sensor **122**, for example, consists of a CMOS image sensor and its peripheral circuitry. The CMOS image sensor has a photoelectric conversion element in each pixel that receives light, and has a pixel group (imaging plane) in which a plurality of unit pixels are arranged in a two-dimensional array, with each pixel as a unit pixel. The image sensor **122** has a plurality of focus detection pixels that receive light beams passing through different pupil areas of the image pickup optical system, respectively, and is capable of output signals independently for each pixel. This enables detection (calculation) of the defocus amount by the imaging plane phase-difference AF. The image sensor **122** has a plurality of image pickup pixels that generate an image signal of an object by receiving light beams passing through an entire area of the exit pupil of the image pickup optical system that forms an image of the object.

(33) The drive/control system of the camera body **120** has an image sensor driver **123**, an image processor **124**, the camera MPU **125**, a display unit **126**, an operation switch **127**, a memory **128**, and a phase-difference AF unit **129**. The drive/control system of the camera body **120** also has an AE unit **130**, a white balance adjuster **131**, and an object detector **132**.

(34) The image sensor driver **123** controls a charge accumulation operation of the image sensor **122**, converts an image signal read out from the image sensor **122** into a digital signal and sends the digital signal to the camera MPU **125**. The image processor **124** performs various image processing such as a gamma conversion, a color interpolation, and a JPEG compression on the image signal read out from the image sensor **122**. The image processor **124** also generates a signal for the imaging plane phase-difference AF described later (focus detection signal), a signal for an exposure adjustment, a signal for white balance adjustment, and a signal for an object detection.

(35) The camera MPU **125** as a controller is a computer having at least one microprocessor. The camera MPU **125** performs operations related to the camera body **120** and controls the image sensor driver **123**, the image processor **124**, the display unit **126**, the operation switch **127**, the memory **128**, and the phase-difference AF unit **129**. The camera MPU **125** can communicate with the lens MPU **117** via signal lines disposed on the mount M. This causes the camera MPU **125** to issue, to the lens MPU **117**, a request to acquire a lens position, a request for a zoom drive, a diaphragm drive, or a lens drive at a predetermined drive amount, or a request to acquire optical information specific to the lens unit **100**.

(36) The camera MPU **125** contains a ROM **125a** that stores a program for controlling camera operations, a RAM **125b** that stores variables, and an EEPROM **125c** that stores various parameters. The camera MPU **125** reads the program stored in the ROM **125a**, expands it in the RAM **125b**, and executes a focus adjustment process, an object detection process, an exposure adjustment process, and a white balance adjustment process according to the program.

(37) The display unit **126** has a display device such as an LCD (liquid crystal) panel or an organic EL, and displays various information about an operation mode set in the camera body **120**. The operation mode includes a still image capturing mode, a moving image capturing mode, and a playback mode, which plays back captured images stored in the memory **128**.

(38) The operation switch **127** includes a shutter switch, a power switch, a zoom switch, a mode switching switch, and a search switch (instruction means). The memory **128** is a flash memory that can be attached to or detached from the camera and records captured images.

(39) The phase-difference AF unit **129** as a focus detector performs a focus detection using the imaging plane phase-difference detecting method based on a focus detection signal as a pair of image signals with parallax to each other for the focus detection obtained from the image sensor **122** and the image processor **124**. Specifically, the image processor **124** calculates an image shift amount (phase difference) between a pair of phase-difference image data generated from the pair of

focus detection signals by performing a correlation calculation on the pair of phase-difference image data. Then, the image processor **124** converts the image shift amount into a defocus amount to detect the defocus amount. The phase-difference AF unit **129** performs a focus adjustment (AF) process that controls a position of the focus lens **104** using the detected defocus amount (focus detection result). The phase-difference AF unit **129** may perform a focus detection using a phase-difference detecting method with a focus detection sensor separate from the image sensor **122**, instead of the imaging plane phase-difference detecting method.

(40) The phase-difference AF unit **129** in this embodiment has a signal generation block **129a** that generates first and second focus detection signals described below and a calculation block **129b** that calculates a phase difference between the first and second focus detection signals and also calculates a defocus amount from the phase difference. At least part of the phase-difference AF unit **129** (part of the signal generation block **129a** or calculation block **129b**) may be provided in the camera MPU **125**. An AF process (focus control process) executed by the camera MPU **125** and the phase-difference AF unit **129** is described later. The camera MPU **125** and the phase-difference AF unit **129** constitute a focus control unit.

(41) The object detector **132** performs an object detection process to detect a type, part, and state of an object (detection type), and a position and size of the object (detection area), based on a signal for an object detection generated by the image processor **124**.

(42) The AE unit **130** controls an exposure condition by performing photometry based on a signal for an exposure adjustment obtained from the image sensor **122** and the image processor **124**. Specifically, an exposure amount at a currently set aperture value, shutter speed, and ISO sensitivity is calculated, and an appropriate aperture value, shutter speed, and ISO sensitivity for image capturing are calculated based on a difference between the calculated exposure amount and a predetermined appropriate exposure amount, and are set as the exposure condition. This provides an automatic exposure adjustment (AE).

(43) The white balance adjuster **131** performs a white balance adjustment process based on a signal for a white balance adjustment obtained from the image sensor **122** and the image processor **124**. Specifically, the white balance adjuster **131** adjusts a color weighting based on a difference between a white balance parameter obtained from the signal for the white balance adjustment and a predetermined appropriate white balance parameter. This provides an automatic white balance adjustment (AWB).

(44) The camera body **120** in this embodiment can perform AF, AE, and AWB in combination with the object detection, and can select a position where AF, AE, and AWB are performed in a range to be captured according to the object detection result.

(45) <Image Sensor Configuration>

(46) FIG. 2 shows an arrangement of image pickup pixels in the image sensor **122** as a two-dimensional CMOS sensor in a range of 4 columns by 4 rows, and an arrangement of focus detection pixels in a range of 8 columns by 4 rows. In an image pickup pixel group of 2 columns×2 rows shown in FIG. 2, the image pickup pixel **200R** with R (red) spectral sensitivity is located in the upper left, the image pickup pixels **200G** with G (green) spectral sensitivity are located in the upper right and lower left, and an image pickup pixel **200B** with B (blue) spectral sensitivity is located in the lower right. Furthermore, each image pickup pixel is composed of a first focus detection pixel **201** and a second focus detection pixel **202** arranged in two rows by one column. By arranging a large number of such image pickup pixel groups **200** on the imaging plane, it is possible to acquire a captured image and a focus detection signal.

(47) FIG. 3A illustrates one image pickup pixel (hereinafter simply called pixel) **200G** of the image sensor **122** shown in FIG. 2, viewed from a light-receiving side (+z side) of the image sensor **122**. FIG. 3B illustrates the a-a cross section of FIG. 3A from the -y side.

(48) The pixel **200G** has a microlens **305** for condensing incident light and a photoelectric converter **301** and a photoelectric converter **302**, which are divided into two sections in the x

direction. The photoelectric converter **301** and the photoelectric converter **302** correspond to the first focus detection pixel **201** and the second focus detection pixel **202**, respectively, shown in FIG. 2.

(49) The photoelectric converter **301** and the photoelectric converter **302** may be pin-structured photodiodes with an intrinsic layer between a p-type layer and an n-type layer, or they may be pn-junction photodiodes with the intrinsic layer omitted. The pixel **200G** has a color filter **306** between the microlens **305** and the photoelectric converters **301** and **302**. A spectral transmittance of the color filter may be changed for each photoelectric converter, or the color filter may be omitted.

(50) Light incident on the pixel **200G** is condensed by the microlens **305**, spectrally split by the color filter **306**, and then received by the photoelectric converter **301** and the photoelectric converter **302**. In the photoelectric converter **301** and the photoelectric converter **302**, pairs of electrons and holes are generated according to received light amount, and after being separated by a depletion layer, the negatively charged electrons are accumulated in the n-type layer, while the holes are discharged to the outside of the image sensor **122** through the p-type layer connected to a constant voltage source (not shown).

(51) The electrons accumulated in the n-type layers of the photoelectric converter **301** and the photoelectric converter **302** are transferred to a capacitance unit (FD) through a transfer gate and converted into a voltage signal.

(52) FIG. 4 illustrates the correspondence between the pixel structure of the image sensor **122** shown in FIGS. 3A and 3B and a pupil division. FIG. 4 shows a cross section of the pixel structure of the image sensor **122** shown in FIG. 3A from the ty side and a pupil plane of the image sensor **122** (pupil distance D_s). In FIG. 4, the x- and y-axes of the cross section of the image sensor **122** are shown inverted with respect to FIGS. 3A and 3B to correspond with the coordinate axes of the pupil plane of the image sensor **122**.

(53) In FIG. 4, a first pupil partial area **501** is an area in which the first focus detection pixel **201** can receive light, which is approximately conjugated by the microlens **305** to a light receiving surface of the photoelectric converter **301** whose center of gravity is decentered in the $-x$ direction. A second pupil partial area **502** is an area in which the second focus detection pixel **202** can receive light, which is approximately conjugated by the microlens **305** to a light receiving surface of the photoelectric converter **302** whose center of gravity is decentered in the $+x$ direction. In FIG. 4, a pupil area **500** including the first pupil partial area **501** and the second pupil partial area **502** is an area in which the entire pixel **200G** combined with the photoelectric converters **301** and **302** (first and second focus detection pixels **201** and **202**) can receive light.

(54) As shown in FIG. 5, light beams passing through the first pupil partial area **501** and the second pupil partial area **502**, which are mutually different, in the pupil area **500** of the image pickup optical system enter the respective pixels on the imaging plane **800** at angles different from each other and are received by the first focus detection pixel **201** and the second focus detection pixel **202**, respectively. FIG. 5 shows an example where the pupil area is divided into two pupils in a horizontal direction, but it may also be divided vertically. The light beams passing through the pupil area **500** are received by the first focus detection pixel **201** and the second focus detection pixel **202** in the respective pixels.

(55) A first focus detection signal is generated by combining photoelectric conversion signals from the first focus detection pixels **201** of multiple pixels, and a second focus detection signal is generated by combining photoelectric conversion signals from the second focus detection pixels **202**. In each pixel, the photoelectric conversion signals from the first and second focus detection pixels **201** and **202** are added and the photoelectric conversion signals from all pixels are combined to generate an image pickup signal with a resolution of N effective pixels. The second focus detection signal may be generated by subtracting the first focus detection signal from the image pickup signal.

(56) <Relationship Between Defocus Amount and Image Shift Amount>

(57) FIG. 6 illustrates the relationship between the defocus amount and the image shift amount between the first and second focus detection signals. As shown in FIG. 5, the pupil area of the image pickup optical system is divided into two pupil areas: the first pupil partial area **501** and the second pupil partial area **502**. The defocus amount d is defined as a distance from an imaging position of the object image to the imaging plane **800**, and a front focus state in which the imaging position of the object image is closer to the object than the imaging plane **800** is indicated by a negative sign ($d < 0$). A positive sign ($d > 0$) indicates a rear focus state in which the imaging position of the object image is farther from the object than the imaging plane **800**. In an in-focus state in which the imaging position of the object image is on the imaging plane **800**, $d = 0$. In FIG. 6, an object **801** indicates an object in the in-focus state ($d = 0$), and an object **802** indicates an object in the front focus state ($d < 0$). The front focus state ($d < 0$) and the rear focus state ($d > 0$) together are called a defocus state ($|d| > 0$).

(58) In the front focus state ($d < 0$), among the light beams from the object **802**, the light beams that have passed through the first and second pupil partial areas **501** and **502** are condensed once and then spread to widths Γ_1 and Γ_2 centered on the center of gravity positions G_1 and G_2 of the light beams, respectively, resulting in a blurred image on the imaging plane **800**. The blurred image is received by the first and second focus detection pixels **201** and **202**, which generate the first and second focus detection signals. Therefore, the first and second focus detection signals are recorded as object images in which the object **802** is blurred with the widths Γ_1 and Γ_2 at the center of gravity positions G_1 and G_2 on the imaging plane **800**, respectively. The blur widths Γ_1 and Γ_2 of the object image generally increase roughly proportionally as the magnitude $|d|$ of the defocus amount d increases. Similarly, the magnitude $|p|$ of the image shift amount p between the first and second focus detection signals (difference $G_1 - G_2$ in the center of gravity positions of the light beams), also increases roughly proportionally as the magnitude $|d|$ of the defocus amount d increases. The same is true for the rear focus state ($d > 0$), except that an image shift direction between the first and second focus detection signals is opposite to that in the front focus state.

(59) The phase-difference AF unit **129** converts, based on a relationship that the image shift amount between the first and second focus detection signals increases as the defocus amount increases, the image shift amount to the defocus amount d by a conversion coefficient calculated based on the distance (baseline length) between the first and second focus detection pixels **201** and **202**.

(60) <Image Pickup Process>

(61) The flowchart in FIG. 7 shows the image pickup process that the camera MPU **125** executes according to the program in this embodiment.

(62) In the step **S701**, the camera MPU **125** causes the phase-difference AF unit **129** to perform the focus detection and acquires the defocus amount as the focus detection result.

(63) Next, in the step **S702**, the camera MPU **125** determines whether or not an AF instruction is received, and in a case where the AF instruction is received, the flow proceeds to the step **S703**; in a case where the AF instruction is not received, the flow proceeds to the step **S704**.

(64) In the step **S703**, the camera MPU **125** executes a normal AF (imaging plane phase-difference AF) process and sets a drive amount of the focus lens **104** (hereinafter referred to as focus drive amount) according to the defocus amount acquired in the step **S701**. Then, the flow proceeds to the step **S706**.

(65) In the step **S704**, the camera MPU **125** determines whether or not a search instruction is given by a user's operation of the search switch of the operation switch **127**. The search instruction includes an instruction to start a search AF process and an instruction for a search direction (drive direction of the focus lens **104**). In a case where the search instruction is given, the flow proceeds to the step **S705**, and in a case where the search instruction is not given, the flow proceeds to the step **S701**.

(66) In the step **S705**, the camera MPU **125** executes the search AF process and then the flow proceeds to the step **S706**. The search AF process is described later.

(67) In the step **S706**, the camera MPU **125** sends the focus drive amount set in the step **S703** or the step **S705** to the lens MPU **117** to drive the focus lens **104**.

(68) In the step **S707**, the camera MPU **125** determines whether or not the image pickup optical system is focused on the object, and in a case where the camera MPU **125** determines that the object is in focus, the flow proceeds to the step **S708**, and in a case where the camera MPU **125** determines that the object is not in focus, the flow proceeds to the step **S701**.

(69) In the step **S708**, the camera MPU **125** executes image capturing for recording. When the image capturing is completed, this process is ended.

(70) <Search AF Process>

(71) The flowchart in FIG. **8** shows the search AF process (focus control method) executed in the step **S705**. In the search AF (process), a search operation (hereinafter simply referred to as search) is performed to perform the focus detection at a predetermined cycle while moving the focus lens **104** to search for an in-focus position of the focus lens **104**. The focus lens **104** is then moved to the in-focus position specified by the search.

(72) In the step **S801**, the camera MPU **125** acquires the search direction in the search instruction determined in the step **S704** of FIG. **7**.

(73) Next, in the step **S802**, the camera MPU **125** acquires a current position (search start position) of the focus lens **104** from the lens MPU **117**.

(74) Next, in the step **S803**, the camera MPU **125** calculates a movable range of the focus lens **104** (hereinafter referred to as focus movable range). The calculation of the focus movable range is described later.

(75) Next, in the step **S804**, the camera MPU **125** determines whether or not the defocus amount acquired in the step **S701** of FIG. **7** is a defocus amount for a position within the focus movable range calculated in the step **S803**. In other words, the camera MPU **125** determines whether or not a target position of the focus lens **104** is within the focus movable range in a case where the focus lens **104** is driven by the focus drive amount based on the defocus amount. In a case where the defocus amount for the position within the focus movable range is acquired (the target position is within the focus movable range), the flow proceeds to the step **S805**, and in a case where the defocus amount is not a defocus amount for the position within the focus movable range, the flow proceeds to the step **S806**. Here, in a case where the defocus amount is not reliable enough to be used, it is determined that the defocus amount is not within the focus movable range.

(76) In the step **S805**, the camera MPU **125** sets the focus drive amount based on the defocus amount acquired in the step **S701**. Then, this process is ended.

(77) In the step **S806**, the camera MPU **125** sets a predetermined focus drive amount in the search direction acquired in the step **S801**, without using the defocus amount acquired in the step **S701**. The predetermined focus drive amount is a drive amount during the search, and may be set according to a cycle of the focus detection, an image pickup distance, an aperture value, an image pickup mode, and the like. After the step **S806**, the process is ended.

(78) <Calculation of Focus Movable Range>

(79) The calculation of the focus movable range in the step **S803** of FIG. **8** is explained. FIG. **10** shows a positional relationship between the object and the background. The object is located at a position close to the image pickup system **10** and the background is located far enough away from the image pickup system **10**.

(80) FIGS. **11A** and **11B** show a signal indicating the object (hereinafter referred to as “object signal”) and a signal indicating the background (hereinafter referred to as “background signal”), these signals being acquired from the image sensor **122** when the object and the background are in the positional relationship shown in FIG. **10**. FIG. **11A** shows the object signal **1102** and the background signal **1101** when the image pickup optical system is focused on the background. FIG. **11B** shows the object signal **1104** and the background signal **1103** when the image pickup optical system is focused on the object. In reality, the object signal and the background signal are acquired

from the image sensor **122** as a signal in which they are added, but here the object signal and the background signal are shown separately.

(81) In a background-focused state shown in FIG. **11A**, a contrast of the background signal **1101** is high and a contrast of the object signal **1102** is very low. Therefore, the background signal **1101** greatly affects the background-focused state, and the defocus amount relative to the background is detected as the focus detection result. On the other hand, in an object-focused state shown in FIG. **11B**, a contrast of the object signal **1104** is high and a contrast of the background signal **1103** is very low. Therefore, the object signal **1104** greatly affects the object-focused state, and the defocus amount relative to the object is detected as the focus detection result.

(82) FIG. **12** shows a relationship between the focus lens position and the focus detection result when the object and the background are in the positional relationship shown in FIG. **10**. A horizontal axis indicates the focus lens position and a vertical axis indicates the defocus amount. The search direction is a direction from a long (background) side to a short (object) side of the in-focus image pickup distance.

(83) When the focus lens position is in the vicinity of a background-focused position (**1201**, **1202**), the defocus amount relative to the background is detected, which is greatly affected by the background signal as described above. On the other hand, when the focus lens position is in the vicinity of an object-focused position (**1203**, **1204**), the defocus amount relative to the object is detected, which is greatly affected by the object signal. In a section between the vicinity of the background-focused position and the vicinity of the object-focused position, both the background signal and the object signal have low contrast, so that a reliability of the defocus amount is low and the defocus amount that can be used for AF cannot be detected.

(84) The flowchart in FIG. **9** shows a calculation process of the focus movable range. In the step **S901**, the camera MPU **125** calculates a difference x between the search start position of the focus lens **104** acquired in the step **S802** at the start of the search AF (search start) and the current position of the focus lens **104** acquired in the step **S802** at the current frame.

(85) Next, in the step **S902**, the camera MPU **125** determines whether or not the difference x is less than or equal to a predetermined first threshold value $Th1$, and in a case where the difference x is less than or equal to the first threshold value $Th1$, the flow proceeds to the step **S903**, and in a case where the difference x is larger than the first threshold value $Th1$, the flow proceeds to the step **S905**.

(86) In the step **S903**, the camera MPU **125** acquires a focus detectable range of the phase-difference AF unit **129**, and further acquires the aperture value and focus sensitivity (optical information of the image pickup optical system) being set from the lens MPU **117**. The focus detectable range is an image blur amount (spread amount of the object image) detectable by the phase-difference AF unit **129**. The focus sensitivity indicates a relationship (ratio) between a unit drive amount of the focus lens **104** and an amount of change in the defocus amount.

(87) Next, in the step **S904**, the camera MPU **125** calculates an offset amount in the same direction as the search direction acquired in the step **S801**, based on the difference x calculated in the step **S901** and the focus detectable range R , aperture value F and focus sensitivity S acquired in the step **S903**. The offset amount is a drive amount of the focus lens **104** calculated by considering a case where the search start position is in a direction opposite to the search direction relative to the background-focused position, and is calculated, for example, by the following formula (1). Here, α indicates a predetermined gain value.

Offset amount = $\alpha(R/x)FS$

(88) When an image blur amount exceeding the focus detectable range occurs for an object in the vicinity of the search start position, the focus detection result for the object cannot be obtained, so that there is no need to set an offset amount that causes the image blur amount exceeding the focus detectable range. Therefore, in such cases, the offset amount is set based on the focus detectable range.

(89) As the current position of the focus lens **104** moves away from the search start position, the possibility that the current position exceeds the background-focused position increases. For this reason, by decreasing the offset amount as the current position moves away from the search start position so as to be inversely proportional to the difference x between the search start position and the current position, an excessively large offset amount is prevented from being set. The aperture value F is used for conversion from the image blur amount to the defocus amount, and the focus sensitivity is used for conversion from the defocus amount to the focus drive amount. The formula (1) is an example of a formula for calculating the offset amount, and the offset amount may be calculated by other methods. From the step **S904**, the flow proceeds to the step **S909**.

(90) In the step **S909**, the camera MPU **125** calculates the focus movable range based on the current position of the focus lens **104**, the offset amount calculated in the step **S904**, and the search direction acquired in the step **S801**. The focus movable range is a range from a position shifted (separated) from the current position in the search direction by the offset amount to a drive end (control end or mechanical end) of the focus lens **104** in the search direction. The camera MPU **125** that has calculated the focus movable range ends this process.

(91) FIGS. **13A**, **13B**, **13C**, and **13D** each illustrates a relationship among the search direction, the search start position, the focus lens position, the focus movable range, the defocus amount, and the focus drive amount. The focus positions (current positions) **1201-1204** in the figures correspond to the positions **1201-1204** shown in FIG. **12**.

(92) FIG. **13A** shows a state at the search start, where the focus lens **104** is positioned at the search start position as the current position **1201**. Since $x(=0) \leq Th1$ in this state, the offset amount **1302a** is set in the same direction as the search direction based on the focus detectable range, the aperture value, the focus sensitivity and x (step **S904**). The focus movable range **1303a** is set in the search direction from a position shifted in the search direction from the search start position by the offset amount **1302a**. In this state, the defocus amount **130a** up to the background-focused position is detected. However, the background-focused position as the target position of the focus lens **104** based on the defocus amount **130a** is outside the focus movable range **1303a**. For this reason, the defocus amount **130a** is not used, and a predetermined amount for the search is set as the focus drive amount **1304a** (step **S806**).

(93) In this way, the focus movable range shifted by the offset amount in the search direction from the search start position (current position) is set. As a result, even if the search start position is located in the direction opposite to the search direction relative to the background-focused position, the object-focused position can be searched without focusing on the background.

(94) FIG. **13B** shows a state in which the focus lens **104** has moved from the search start position to the current position **1202** closer to the object-focused position than the background-focused position. Since $x(>0) \leq Th1$ in this state as well, the offset amount **1302b** is set in the same direction as the search direction based on the focus detectable range, the aperture value, the focus sensitivity, and x (step **S904**). The focus movable range **1303b** is set in the search direction from a position shifted from the current position **1202** by the offset amount **1302b**. Since x is larger than the state in FIG. **13A**, the offset amount **1302b** is smaller than the offset amount **1302a**. Even in this state, the defocus amount **1301b** up to the background-focused position is detected. However, the background-focused position as the target position of the focus lens **104** based on the defocus amount **1301b** is outside the focus movable range **1303b**. For this reason, the defocus amount **1301b** is not used, and a predetermined amount for the search is set as the focus drive amount **1304b** (step **S806**). Although the focus drive amount **1304b** is the same as the focus drive amount **1304a** shown in FIG. **13A**, the focus drive amount **1304b** may be smaller than or different from the focus drive amount **1304a**.

(95) In this way, the focus movable range shifted by the offset amount from the focus lens position after the search start is set. This makes it possible to search for the object-focused position without focusing on the background, even if the detected defocus amount **1301a** is for a position within the

focus movable range (**1303a**) set at the search start.

(96) In the step **S905** in FIG. **9**, the camera MPU **125** determines whether or not the difference x is greater than or equal to a predetermined second threshold value $Th2$ ($>Th1$), and in a case where the difference x is greater than or equal to the threshold value $Th2$, the flow proceeds to the step **S906**, and in a case where the difference x is smaller than the threshold value $Th2$, the flow proceeds to the step **S908**.

(97) In the step **S906**, the camera MPU **125** acquires a drive velocity of the focus lens **104** and a focus detection cycle.

(98) Next, in the step **S907**, the camera MPU **125** calculates the offset amount in the direction opposite to the search direction acquired in the step **S801** based on the drive velocity and focus detection cycle acquired in the step **S906**, as well as the difference x calculated in the step **S901**. The offset amount here is a relationship between the drive velocity v of the focus lens **104** and the focus detection cycle T , and is set in consideration of the case where the object-focused position is overshoot during the search, and is calculated, for example, by the following formula (2). β is a predetermined gain value.

$$\text{Offset amount} = \beta v T x$$

(99) The drive amount of the focus lens **104** between frames in which the focus detection is performed is calculated by the product of the drive velocity v and the focus detection cycle T . Since the drive amount of the focus lens **104** between the above frames is the maximum amount by which the object-focused position is overshoot, the offset amount is set based on this drive amount. As the position of the focus lens **104** is far away from the search start position, the possibility that the position of the focus lens **104** overshoots the object-focused position becomes high. For this reason, by increasing the offset amount in proportion to the difference x between the search start position and the current position of the focus lens **104**, the object-focused position can be easily captured within the movable range even in the case of overshooting. That is, since the possibility of overshooting at the search start is low, the movable range is narrowed to reduce the risk of returning to the background. When the position of the focus lens **104** is far away from the search start position, the risk of returning to the background decreases and the risk of overshooting increases. For this reason, by widening the movable range in the search start direction, it is easy to bring the object-focused position within the drivable range even in the case of overshooting. The method for calculating the offset amount described above is merely an example, and other methods may be used for calculation. For example, the offset amount (that is, the focus movable range) may be set based on only one of the drive velocity of the focus lens **104** and the focus detection cycle.

(100) On the other hand, in the step **S908**, the camera MPU **125** sets the offset amount to zero. Then, the flow proceeds to the step **S909** to calculate the focus movable range as described above, and the process is ended.

(101) FIG. **13C** shows a state in which the focus lens **104** has moved to the current position **1203** closer to the object-focused position than in the state in FIG. **13B**. In this state, since $x \geq Th2$, the offset amount **1302c** is set in the direction opposite to the search direction based on the drive velocity of the focus lens **104**, the focus detection cycle, and the difference x (step **S907**). The focus movable range **1303c** is set in the search direction from a position shifted from the current position **1203** by the offset amount **1302c** in the direction opposite to the search direction. In this state, the defocus amount **1301c** up to the object-focused position is detected, and the object-focused position as the target position of the focus lens **104** based on the defocus amount **1301c** is within the focus movable range **1303c**. For this reason, the focus drive amount **1304c** is set based on the defocus amount **1301c** (step **S805**).

(102) In this way, the focus movable range is set in the search direction from the position shifted by the offset amount in the direction opposite to the search direction from the current position of the focus lens **104**. This makes it possible to focus on the object by driving the focus lens **104** based on the defocus amount detected at a timing when the object-focused position is within the focus

movable range.

(103) FIG. 13D shows a state in which the focus lens **104** has moved to the current position **1204** that has overshoot the object-focused position. Even in this state, since $x \geq Th2$, the offset amount **1302d** is set in the direction opposite to the search direction based on the drive velocity of the focus lens **104**, the focus detection cycle, and the difference x (step **S907**). The focus movable range **1303d** is set in the search direction from a position shifted from the current position **1204** by the offset amount **1302d** in the direction opposite to the search direction. Since the difference x is larger than the state in FIG. 13C, the offset amount **1302d** is larger than the offset amount **1302c**. In this state, the defocus amount **1301d** up to the object-focused position located in the opposite direction to the search direction is detected, and the object-focused position as the target position of the focus lens **104** based on the defocus amount **1301d** is within the focus movable range **1303d**. For this reason, the focus drive amount **1304d** is set based on the defocus amount **1301d** (step **S806**).

(104) In this way, the focus movable range is set in the search direction from the position shifted by the offset amount in the direction opposite to the search direction from the current position of the focus lens **104**. As a result, even when the focus lens **104** overshoots the object-focused position in the search direction, the focus lens **104** can be driven based on the defocus amount to focus on the object.

(105) As described above, in this embodiment, the focus movable range is set from the position shifted from the current position of the focus lens **104** by the offset amount. Then, the focus drive amount is set according to whether or not the focus detection result for the position within the focus movable range is acquired. As a result, an appropriate search AF can be performed for an object desired by the user, and the in-focus state for the object can be quickly acquired.

Second Embodiment

(106) Next, the second embodiment will be explained. In the second embodiment, a direction of the defocus amount as the focus detection result (hereinafter referred to as “defocus direction”) is determined, and the drive amount of the focus lens is set according to whether or not the defocus direction is the same direction as the search direction, and the search is performed.

(107) The configuration of the image pickup system **10** and the image sensor **122** and the image pickup process in the second embodiment are the same as those in the first embodiment. The positional relationship between the object and the background, the relationship between the object signal and the background signal, as well as the relationship between the focus lens position and the focus detection result are the same as those in the first embodiment.

(108) <Search AF Process>

(109) FIG. 14 shows the search AF process executed in the step **S705** in FIG. 7 in this embodiment. In the step **S1401**, the camera MPU **125** acquires the search direction in the search instruction determined in the step **S704** in FIG. 7.

(110) Next, in the step **S1402**, the camera MPU **125** determines the defocus direction of the focus detection result acquired in the step **S701**.

(111) Next, in the step **S1403**, the camera MPU **125** calculates the difference x between the search start position of the focus lens **104** acquired in the step **S802** at the search start and the current position of the focus lens **104** acquired in the step **S802** in the current frame.

(112) Next, in the step **S1404**, the camera MPU **125** determines whether or not the difference x is smaller than or equal to a third threshold value $Th3$. In a case where the difference x is less than or equal to the third threshold $Th3$, the flow proceeds to the step **S1408**, and in a case where the difference x is larger than the third threshold $Th3$, the flow proceeds to the step **S1405**.

(113) In the step **S1405**, the camera MPU **125** determines whether or not the search direction and the defocus direction were in the same direction in the past (e.g., last) frame in the range of $x \geq Th3$ (frame for which a reliable defocus amount could be acquired). In a case where they are in the same direction, the flow proceeds to the step **S1407**, and in a case where they are not in the same

direction, the flow proceeds to the step **S1406**. This determination takes into consideration the case where the focus lens **104** overshoots the object-focused position during the search. That is, in a case where the search direction and the defocus direction are in the same direction in the past frame (see FIG. **15C** described later) and the search direction and the defocus direction are opposite in the current frame (see FIG. **15D**), then the object-focused position is considered as overshooting and the flow proceeds to the step **S1407**. Also, in a case where the search direction and the defocus direction are in the same direction in the past frame and the search direction and the defocus direction are in the same direction in the current frame (the case where the state shown in FIG. **15C** continues), the flow proceeds to the step **S1407**.

(114) In the step **S1406**, the camera MPU **125** determines whether or not the search direction acquired in the step **S1401** and the defocus direction determined in the step **S1401** are the same in the current frame. In a case where they are in the same direction, the flow proceeds to the step **S1407**, and in a case where they are not in the same direction, the flow proceeds to the step **S1408**. That is, in a case where the search direction and the defocus direction are opposite in the past frame (see FIG. **15B**) and the search direction and the defocus direction are the same in the current frame (see FIG. **15C**), the flow proceeds to the step **S1407**. Also, in a case where the search direction and the defocus direction are opposite in the past frame and the search direction and the defocus direction are opposite in the current frame (the case where the state shown in FIG. **15B** continues), the flow proceeds to the step **S1408**.

(115) In the step **S1407**, the camera MPU **125** sets the focus drive amount based on the defocus amount of the current frame acquired in the step **S701**. Then, this process is ended.

(116) On the other hand, in the step **S1408**, the camera MPU **125** sets a predetermined focus drive amount in the search direction acquired in the step **S801** without using the defocus amount acquired in the step **S701**. The predetermined focus drive amount is a drive amount for the search as in the first embodiment, and may be set according to the focus detection cycle, the image pickup distance, the aperture value, the image pickup mode, and the like. Then, the camera MPU **125** ends this process.

(117) FIGS. **15A-15D** each shows the relationship among the search direction, the search start position, the focus lens position, the focus movable range, the defocus amount, and the focus drive amount in this embodiment.

(118) FIG. **15A** shows a state at the search start, where the focus lens **104** is positioned at the search start as the current position **1201**. Since $x(=0) \leq Th3$ in this state, the defocus amount **1501a** to the background-focused position is detected, but the defocus amount **150a** is not used and a predetermined amount for the search is set as the focus drive amount **1504a** (step **S1408**).

(119) In this way, in a case where $x \leq Th3$, the focus drive amount for the search is set without using the detected defocus amount. This makes it possible to search for the object-focused position without focusing on the background, even if the search start position is located in the direction opposite to the search direction relative to the background-focused position.

(120) FIG. **15B** shows a state in which the focus lens has moved from the search start position to the current position **1202** closer to the object-focused position than the background-focused position. This state indicates a state where $x \geq Th3$ and the search direction and the defocus direction are opposite in each of the past frame and the current frame within the range of $x \geq Th3$ (NO in the steps **S1405** and **S1406**). For this reason, the defocus amount **1501b** up to the background-focused position is detected, but the defocus amount **1501b** is not used, and a predetermined amount for the search is set as the focus drive amount **1504b** (step **S1408**).

(121) In this way, when the search direction and the defocus direction are opposite directions, the focus drive amount for the search is set without using the detected defocus amount. This makes it possible to search for the object-focused position without focusing on the background, even when the defocus amount up to the background-focused position is detected.

(122) FIG. **15C** shows a state in which the focus lens **104** has moved to the current position **1203**

closer to the object-focused position than the state shown in FIG. 15B. In this state, $x \geq Th3$, and in the past frame (FIG. 15B) in the range of the $x \geq Th3$, the search direction and the defocus direction are opposite directions (NO in step S1405), but in the current frame, the search direction and the defocus direction are the same direction (YES in the step S1406). For this reason, the focus drive amount 1504c is set based on the defocus amount 1501c up to the object-focused position detected in the current frame (step S1407).

(123) In this way, when the search direction and the detection direction are the same direction, the focus drive amount is set based on the detected defocus amount. This makes it possible to drive the focus lens 104 to focus on the object based on the defocus amount detected at the timing when the defocus direction becomes the same direction as the search direction.

(124) FIG. 15D shows a state in which the focus lens 104 has moved to the current position 1204, which is past the object-focused position. In this state, $x \geq Th3$, and the search direction and the defocus direction are opposite in the current frame, but the search direction and the defocus direction are the same direction in the past frame (FIG. 15C) in the range of $x \geq Th3$ (Yes in the step S1405). For this reason, the focus drive amount 1504d is set based on the defocus amount 1501d detected in the current frame (step S1407).

(125) In this way, when the search direction and the defocus direction are the same direction in the past frame, the focus drive amount is set based on the detected defocus amount. This makes it possible, even if the focus lens 104 overshoots the object-focused position, to detect the defocus amount up to the object-focused position located in the opposite direction to the search direction, and to drive the focus lens 104 to focus on the object based on the defocus amount.

(126) As described above, according to the second embodiment, the focus drive amount is set according to the determination result of whether or not the search direction and the defocus direction are the same direction. In $x \geq Th3$, after the defocus direction becomes the same direction as the search direction in the past frame, the focus lens 104 is driven based on the defocus amount in the current frame regardless of the defocus direction in the subsequent current frame. As a result, the appropriate search AF can be performed for the object desired by the user, and the in-focus state for the object can be quickly acquired.

(127) In each of the above-described embodiments, the case where the search direction is set in a direction from the long side to the short side of the in-focus image pickup distance to focus on the object while avoiding focusing on the background has been explained. On the other hand, the search AF may be performed in the same manner as in each embodiment by setting the search direction in a direction from the short side to the long side of the image pickup distance. For example, when capturing an image of an animal in a cage, it is possible to focus on the animal in the cage (on the long image pickup distance side) while avoiding focusing on the front of the cage (on the short image pickup distance side).

OTHER EMBODIMENTS

(128) Embodiment(s) of the disclosure can also be realized by a computer of a system or apparatus that reads out and executes computer-executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer-executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer-executable instructions. The computer-executable instructions may be

provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read-only memory (ROM), a storage of distributed computing systems, an optical disc (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

(129) While the disclosure has been described with reference to embodiments, it is to be understood that the disclosure is not limited to the disclosed embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(130) According to the present disclosure, an appropriate search operation can be performed in a focus adjustment using a phase-difference detecting method.

(131) This application claims the benefit of Japanese Patent Application No. 2022-184290, filed on Nov. 17, 2022, which is hereby incorporated by reference herein in its entirety.

Claims

1. A focus control apparatus comprising: a memory configured to store instructions; and a processor configured to execute the instructions, the processor being configured to preform a focus detection using a phase-difference detecting method, and control a drive of a focus lens included in an optical system based on a focus detection result acquired by the focus detection, wherein the processor sets, in a search operation of acquiring the focus detection result while driving the focus lens in a search direction, a movable range of the focus lens based on the search direction and a position of the focus lens, in a case where the focus detection result for a position within the movable range is acquired, drives the focus lens based on the focus detection result, and in a case where the focus detection result for a position outside the movable range is acquired, drives the focus lens in the search direction without using the focus detection result.
2. The focus control apparatus according to claim 1, wherein the processor sets the movable range based on a difference between a start position of the search operation of the focus lens and an actual position of the focus lens.
3. The focus control apparatus according to claim 2, wherein the processor sets the movable range from a position away from the focus lens by an offset amount set based on the difference.
4. The focus control apparatus according to claim 3, wherein the processor sets the offset amount in a same direction as the search direction in a case where the difference is smaller than a first threshold value, and sets the offset amount in a direction opposite to the search direction in a case where the difference is larger than a second threshold value that is larger than the first threshold value.
5. The focus control apparatus according to claim 1, wherein the processor sets the movable range based on a detectable blur amount of an object image.
6. The focus control apparatus according to claim 1, wherein the processor sets the movable range based on optical information on the optical system.
7. The focus control apparatus according to claim 1, wherein the processor sets the movable range based on at least one of a drive velocity of the focus lens or a cycle in which the focus detection is performed.
8. The focus control apparatus according to claim 1, further comprising an instruction means for a user to instruct at least one of a direction in which an in-focus image pickup distance is shortened from a long side or a direction in which the in-focus image pickup distance is lengthened from a short side as the search direction.
9. An image pickup apparatus comprising: a focus control apparatus according to claim 1; and an image sensor configured to capture an image of an object via the optical system.
10. A focus control apparatus comprising: a memory configured to store instructions; and a

processor configured to execute the instructions, the processor being configured to perform a focus detection using a phase-difference detecting method, and control a drive of a focus lens based on a focus detection result acquired by the focus detection, wherein in a search operation of acquiring the focus detection result while driving the focus lens in a search direction, the processor in a case where a difference between a position of the focus lens and a start position of the search operation is larger than a threshold value, and in a case where a direction of the focus detection result is a same direction as the search direction, drives the focus lens based on the focus detection result, and in a case where the direction of the focus detection result is a direction opposite to the search direction, drives the focus lens in the search direction without using the focus detection result.

11. The focus control apparatus according to claim 10, wherein the processor drives the focus lens based on the focus detection result regardless of the direction of the focus detection result after the direction of the focus detection result becomes the same as the search direction after the difference becomes larger than the threshold value.

12. The focus control apparatus according to claim 10, further comprising an instruction means for a user to instruct at least one of a direction in which an in-focus image pickup distance is shortened from a long side or a direction in which the in-focus image pickup distance is lengthened from a short side as the search direction.

13. A focus control method comprising: a step of performing a focus detection using a phase-difference detecting method, and a step of controlling a drive of a focus lens included in an optical system based on a focus detection result acquired by the focus detection, wherein in a search operation of acquiring the focus detection result while driving the focus lens in a search direction, a movable range of the focus lens is set based on the search direction and a position of the focus lens, in a case where the focus detection result for a position within the movable range is acquired, the focus lens is driven based on the focus detection result, and in a case where the focus detection result for a position outside the movable range is acquired, the focus lens is driven in the search direction without using the focus detection result.

14. A non-transitory computer-readable storage medium storing a program that causes a computer to execute a process according to a focus control method according to claim 13.

15. A focus control method comprising: a step of performing a focus detection using a phase-difference detecting method, and a step of controlling a drive of a focus lens based on a focus detection result acquired by the focus detection, wherein in a search operation of acquiring the focus detection result while driving the focus lens in a search direction, in a case where a difference between a position of the focus lens and a start position of the search operation is larger than a threshold value, and in a case where a direction of the focus detection result is a same direction as the search direction, the focus lens is driven based on the focus detection result, and in a case where the direction of the focus detection result is a direction opposite to the search direction, the focus lens is driven in the search direction without using the focus detection result.
